Case Study 7: String-Based Examination Answer Evaluation

Assignment Case No. 31 | Level: Beginner

01 PROBLEM STATEMENT

Compare answer strings against an answer key, calculate correct, incorrect, and unanswered responses, apply negative marking, and produce student-wise results.

Problem interpretation: Evaluate a student's multiple-choice responses against an answer key. Blank answers are unanswered, wrong answers receive negative marks, and the final score is calculated using the supplied policy.

02 OBJECTIVES

Compare each answer with an answer key.

Count correct, incorrect, and unanswered questions.

Apply negative marking to incorrect responses.

Calculate and display the final score.

03 ASSUMPTIONS

Each correct answer earns 4 marks.

Each incorrect answer loses 1 mark.

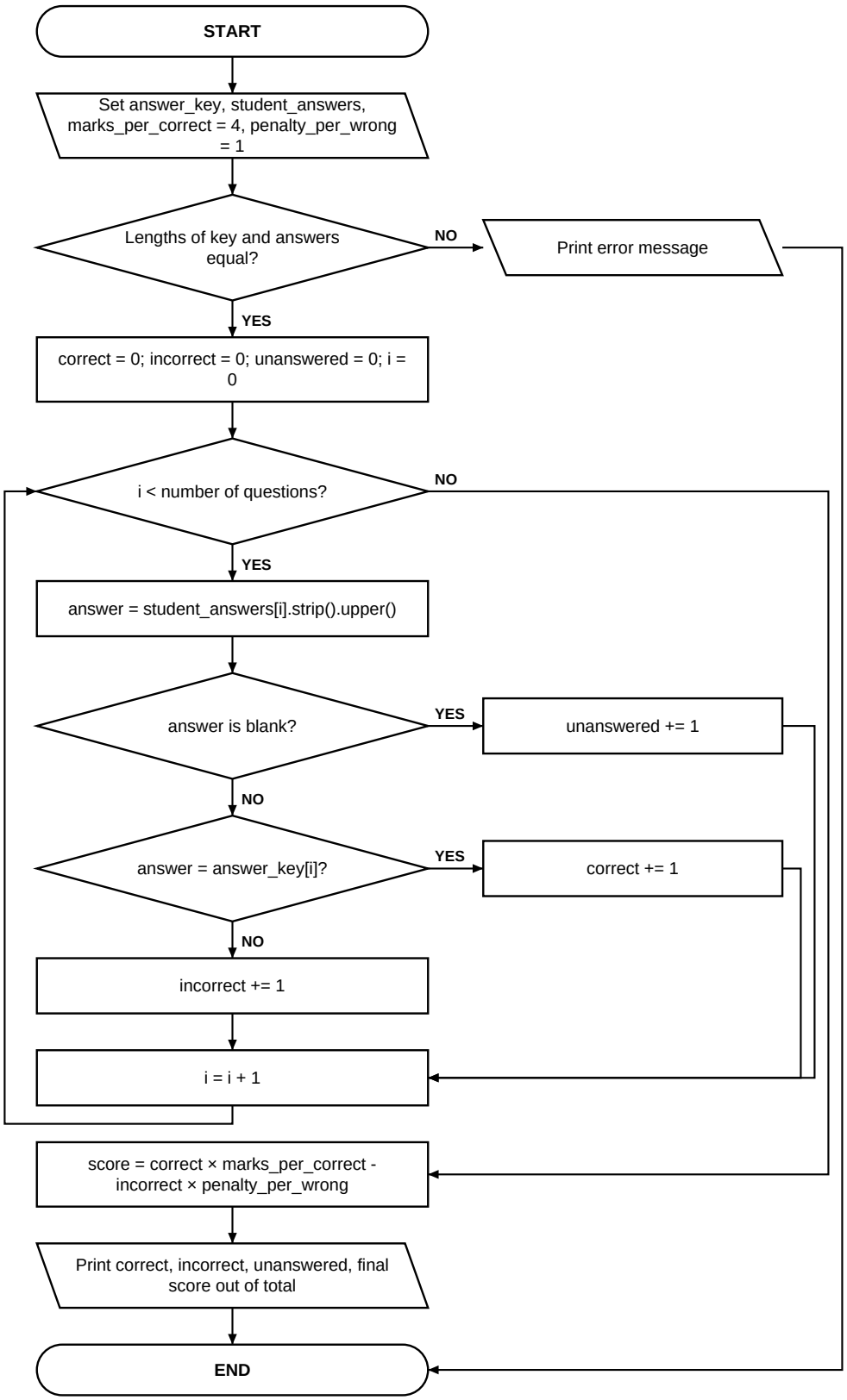
An unanswered question earns 0 marks.

The answer and key lists have the same length.

04 ALGORITHM

1. Start.
2. Store the answer key and student's responses.
3. Initialize correct, incorrect, and unanswered counters.
4. Compare each response with the corresponding key entry.
5. Update the appropriate counter.
6. Calculate score = correct × 4 − incorrect × 1.
7. Display counts and score.
8. Stop.

05 FLOWCHART (TEXT FLOW)



06 PSEUDOCODE

```
BEGIN
    answer_key = [A, C, B, D, A]
    student_answers = [A, B, blank, D, C]
    marks_per_correct = 4; penalty_per_wrong = 1
    IF length(answer_key) != length(student_answers) THEN
        PRINT "Error: answer list and key must have equal length"
    ELSE
        correct = 0; incorrect = 0; unanswered = 0
        FOR i = 0 TO length(answer_key) - 1
            answer = trim(student_answers[i]) in upper case
            IF answer is blank THEN
                unanswered = unanswered + 1
            ELSE IF answer = answer_key[i] THEN
                correct = correct + 1
            ELSE
                incorrect = incorrect + 1
            END IF
        END FOR
        score = correct * marks_per_correct - incorrect * penalty_per_wrong
        PRINT correct, incorrect, unanswered
        PRINT score, total marks = length(answer_key) * marks_per_correct
    END IF
END
```

07 PYTHON SOURCE CODE

The complete Python program is provided in the separate file 07_Examination_Answer_Evaluation_Code.pdf.

08 TEST CASES

INPUT / SCENARIO	CHECK	EXPECTED RESULT
A matches key A	Correct count increases	+4 marks
B given, key is C	Incorrect count increases	-1 mark
Blank response	Unanswered count increases	0 marks
Sample answers vs. key	2 correct, 2 incorrect, 1 unanswered	Final score: 6 out of 20
Key and answers of different length	Length check fails	Error message printed

09 SAMPLE OUTPUT PREVIEW

```
EXAMINATION RESULT
Correct answers: 2
Incorrect answers: 2
Unanswered: 1
Final score: 6 out of 20
```

10 CONCLUSION

The evaluator demonstrates string comparison, list indexing, loops, counters, and a simple negative-marking formula.